Solution Design

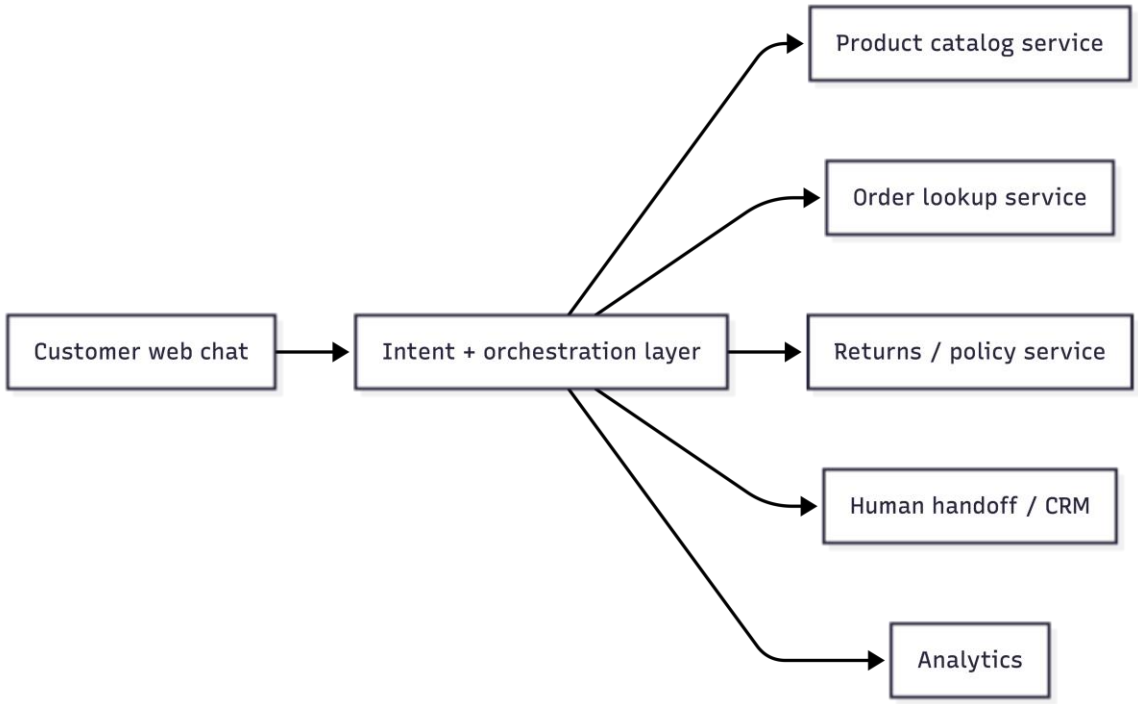
Goal

This POC proves that a focused AI-first support layer can reduce response time for routine e-commerce inquiries without trying to replace the support team. I prioritized the smallest MVP that directly addresses the brief:

- `product information` to support conversion and reduce cart abandonment
- `order tracking` to deflect repetitive tickets
- `returns/refunds` to protect trust after purchase
- `human handoff` to safely handle unclear or sensitive cases

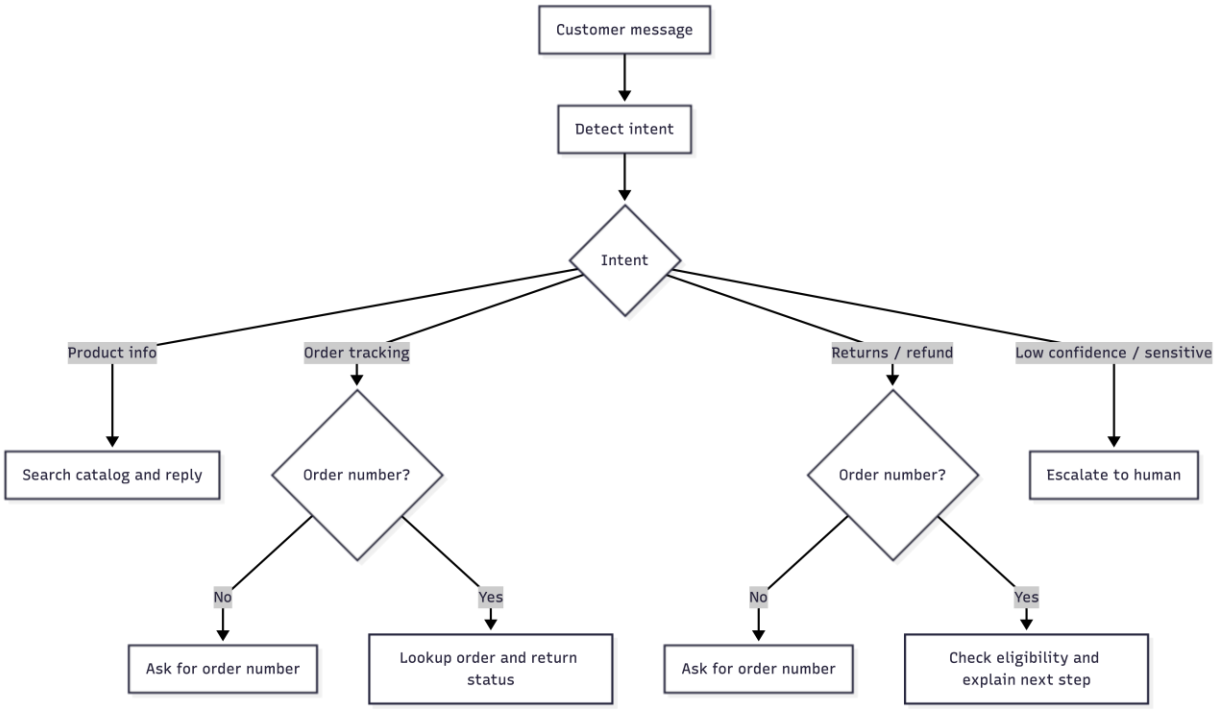
Architecture

The prototype is a React storefront with an embedded support widget. The chatbot sits on top of grounded business services rather than answering free-form generation alone. This keeps product, order, and policy answers trustworthy and makes the shape easy to integrate later with real commerce and CRM systems.



Core user flow

The assistant first detects intent, then follows a grounded path. If key information is missing, it asks a clarifying question. If confidence is low, the issue is sensitive, or the order cannot be resolved safely; it escalates to a human agent.



Key design choices

- `Grounded answers first`: product, order, and refund answers come from structured data, not unsupported generation.
- `Human handoff in MVP`: support systems should not guess on exceptions, complaints, or low-confidence cases.
- `Bilingual-ready`: Arabic and English support matters for KSA operations.
- `Analytics included`: the business needs to measure containment, handoff rate, and response-time improvement.

Technology stack

- frontend: `React`
- backend: `Node.js` native HTTP server
- data layer: shared mock catalog, order, and policy services
- testing: `Node` built-in test runner
- optional AI layer: `OpenAI Responses API` for grounded reply composition when credentials are available